Computational Supplement to “A One-Loop BV Obstruction in Pure Weyl Gravity and Its Formal Tau-Adic Wess–Zumino Resolution”

Asger Alstrup Palm

Working supplement, July 2026

Abstract

This supplement records the exact finite computations underlying the strict one-loop BV obstruction and its formal tau-adic compensator resolution. It separates basis exhaustiveness, relative-cohomology ranks, antifield and gauge-fixing transport, analytic coefficient provenance, the compensator cotangent lift, quartet contraction, the extended quotient, and QME restoration. Tables and matrices carrying numerical or discrete claims are generated directly from content-addressed JSON certificates. Independent verifiers reject stale generated material, altered ranks, missing classes, over-promoted lifecycle states, and dependency-hash drift.

Scope and dependency labels

The supplement proves only the claims labeled LOCAL-ALGEBRAIC and EUCLIDEAN-SPECTRAL in the companion manuscript. It does not prove a Lorentzian QME, renormalized causal products, a Hadamard state, positivity, particles, residual transfer, or an all-loop theorem. The strict and compensator-extended theories remain different rows throughout.

Contents

1 Certificate graph

The calculation is organized as a directed acyclic graph rather than a single symbolic notebook:

$$\begin{array}{c} \text{canonical AFN0 bases and descents} \longrightarrow \text{minimal Koszul–Tate collapse} \\ \downarrow \\ \text{nonminimal/gauge-fixed quotient plus analytic coefficients} \\ \downarrow \\ \text{regulated Slavnov class} \longrightarrow \text{strict QME disposition,} \end{array}$$

and independently

$$\begin{array}{c} \text{compensator master term} \longrightarrow \text{cotangent lift and quartet} \\ \downarrow \\ \text{dressed pure-Diff quotient} \longrightarrow \text{extended QME disposition.} \end{array}$$

No determinant row is promoted directly into BV cohomology. The analytic coefficient vector is accepted only after it is reduced against the complete gauge-fixed quotient with the same normalization and classical snapshot.

2 Exact strict quotient

2.1 Declared local algebra

The production slice is four-dimensional, form degree four, engineering dimension four, and ghost number zero or one. Tensor generators include the metric, curvature and covariant derivatives, the Diff and Weyl ghosts, the minimal antifields, and the declared nonminimal doublets. Epsilon tensors are included in the odd sector. The canonical quotient imposes:

- algebraic and differential Bianchi identities;
- tensor symmetries and four-dimensional antisymmetrization;
- graded factor permutations and Grassmann signs;
- dummy-index graph isomorphism;
- integration by parts and horizontal exactness.

The basis generator is checked twice: forward tensor-graph generation and reverse coverage of every grading-admissible signature. A class receives a complete nontriviality witness only after the top, lower-form cocycle, and lower-form boundary spaces are exhaustive.

2.2 Orbit-first reduction

The even ghost-one basis has seven canonical top carriers before closure and boundary reduction. Orbit-first contraction resolves the two formerly pending mixed signatures by materializing only thirty raw pairings. The odd calculation resolves all three pending mixed signatures: forty-five raw graphs reduce to nine signed orbits, all of which vanish by the Bianchi relations. The single odd Weyl pseudoscalar remains.

The exact rank data and final dimensions are generated below.

Stage	$\dim H^{0,4}_{\text{even}}$	$\dim H^{0,4}_{\text{odd}}$	$\dim H^{1,4}_{\text{even}}$	$\dim H^{1,4}_{\text{odd}}$
strict gauge-fixed BV	2	1	2	1
tau-adic extended BV	3	1	0	0

Table 1: Exact dimension-four local BV quotient dimensions.

Sector	raw graphs/pairings materialized	closure rank	boundary rank
even AFN0 relative carrier	30	6	4
odd AFN0 mixed carrier	45	1	0

Table 2: Orbit-first strict AFN0 quotient audit. The ambient 2.86×10^9 raw-graph expansion is never materialized.

Density coordinate	repository one-loop coefficient
C^2	$\frac{199}{30}$
E_4	$-\frac{87}{20}$
$C\widetilde{C}$	0
$\Box R$	0

Table 3: Coefficient vector in the convention $(4\pi)^{-2}[cC^2 - aE_4 + pC\widetilde{C} + b\Box R]$.

$$Q_W = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad Q_W h + h Q_W = I_4, \tag{1}$$

in the ordered basis $(\tau, \omega, \omega^*, \widehat{\tau}^*)$. The certificate checks 24 atoms and 8 generators over Q; all component gradings, δ^2, $\delta\gamma + \gamma\delta$, and Q^2 pass exactly.

$$B_{\text{ext}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \operatorname{rank} B_{\text{ext}} = 4, \tag{2}$$

from the ordered primitive basis $(B_C, B_E, B_P, B_\Box)$ to $(\omega C^2, \omega E_4, \omega C\widetilde{C}, \omega\Box R)$. The strict breaking and its extended boundary image are both

$$\left(\frac{199}{30}, -\frac{87}{20}, 0, 0\right).$$

Anomaly-induced functional carrier	exact coefficient
$\langle \mathcal{E}_4, G_4 C^2 \rangle$	$\frac{199}{120}$
$\langle \mathcal{E}_4, G_4 \mathcal{E}_4 \rangle$	$-\frac{87}{160}$
$\int \sqrt{g} R^2$	$\frac{29}{120}$

Table 4: Exact Paneitz/Riegert solve. The Weyl-response matrix is $\mathrm{diag}(4, 8, -12)$; re-expansion returns $(199/30, -87/20, 0)$ in $(C^2, E_4, \Box R)$.

Certificate	SHA-256 prefix
`quantum-weyl/local_bv/certificates/AFN0_H14_EVEN_CANONICAL_QUOTIENT.json`	`6bf8b7ef099ce40b`
`quantum-weyl/local_bv/certificates/AFN0_H14_ODD_CANONICAL_QUOTIENT.json`	`0134e28cf3fd0bf4`
`quantum-weyl/local_bv/certificates/GENERAL_NONMINIMAL_GAUGE_FIXED_CONTRACTION.json`	`4513be4824760577`
`quantum-weyl/anomalies/certificates/REGULATED_REPOSITORY_BV_SLAVNOV_BREAKING.json`	`d11ebc782c6a388e`
`quantum-weyl/anomalies/certificates/UNITARY_CONFORMAL_MATTER_CANCELLATION_NO_GO.json`	`9c4b1218450e20f0`
`quantum-weyl/anomalies/certificates/WESS_ZUMINO_MINIMAL_BV_COTANGENT_LIFT.json`	`b265dc9d86938ed7`
`quantum-weyl/anomalies/certificates/WESS_ZUMINO_EXTENDED_LOCAL_BV_COHOMOLOGY.json`	`fa21fc6071ae5227`
`quantum-weyl/transfer/certificates/ANOMALY_INDUCED_NONLOCAL_GAMMA1.json`	`9531ef49d0d930fc`

Table 5: Content-addressed inputs to the generated supplement tables. Full hashes are stored in the Paper 12 claim map.

2.3 Antifields and gauge fixing

The imported minimal classical export supplies every Koszul–Tate row. On the regular Bach locus the positive-antifield sector is an explicit sum of adapted pairs, and its contracting homotopy verifies

$$\delta\sigma + \sigma\delta = N_{\mathrm{pair}}.$$

Consequently the AFN0 Weyl representatives lift through the minimal spectral sequence. A separate total-complex computation proves that the pure-Diff and mixed Diff–Weyl sectors add no independent dimension-four class.

Ten general nonminimal pairs then contract on 25,080 graded regression monomials. Their covariant jet prolongations commute with d_h, and a free-word normal-form computation transports the SDR through an arbitrary invertible local BV-canonical gauge-fixing transformation. Thus the minimal, nonminimal, and gauge-fixed quotient dimensions agree. This explicit chain comparison is why the main theorem is not labeled AFN0-only.

3 Coefficient and Slavnov route

3.1 Analytic input

The accepted Euclidean principal-symbol sequence has dimensions $5 \to 10 \to 5$, ranks $(5, 5)$ at nonzero covector, zero composition, and exact image–kernel equality. The formal adjoint sequence, Diff/Weyl nonminimal doublets, and four kinetic blocks replay independently. The determinant bundle ranks and exponents are

$$(5,1,5,3), \qquad \left(-\frac{1}{2},+\frac{1}{2},-\frac{1}{2},+\frac{1}{2}\right),$$

with priming dimensions $(0,5,0,10)$.

The local C^2 coordinate is exposed on a non-conformally-flat Ricci-flat carrier. The Euler coordinate is independently fixed on round S^4. The measure ledger includes York and ghost super-Jacobians, all nonminimal quartets, conformal-Killing zero modes, the oriented auxiliary contour, and the local parity Ward identity. Global determinant phase and conformal-group volume are not used to infer a local coefficient.

3.2 Cohomological reduction

Let the ordered strict basis be

$$(\omega C^2, \omega E_4, \omega C\widetilde{C}, \omega\Box R).$$

The raw analytic vector is the one displayed in Table 3. The fourth coordinate is removed with the stored R^2 primitive and current. The first two normalized dual functionals evaluate to $199/30$ and $-87/20$, while the odd functional evaluates to zero by the Ward identity. Therefore

$$[\mathcal{A}^{(1)}] = \frac{1}{(4\pi)^2}\left(\frac{199}{30}[\omega C^2] - \frac{87}{20}[\omega E_4]\right) \neq 0.$$

The strict QME verdict is determined by this nonzero cohomology class. The beta function, determinant logarithm, and background trace anomaly are not used as substitutes for the regulated Slavnov insertion.

3.3 Matter cone

For standard-sign free conformal scalars, Weyl/Dirac fermions, and gauge vectors, cancellation is an exact rational cone-feasibility problem. A separating dual covector is positive on every allowed matter ray and on the pure-gravity breaking. Hence no nonnegative real multiplicity vector, and therefore no nonnegative integral multiplicity vector, cancels both even coordinates. This theorem deliberately excludes nonunitary signs, interacting fixed points, conformal higher spin, and compensators.

4 Compensator BV extension

4.1 Rows derived from one master term

The extension is generated from

$$S_{\tau,\min} = \int \tau^*(\xi^\mu\nabla_\mu\tau + \omega).$$

Its four Euler derivatives force

$$\begin{aligned} Q\tau &= \mathcal{L}_\xi\tau + \omega, \\ \delta\omega^* &= N_\omega + \tau^*, \\ \delta\xi^*_\mu &= N_{\xi,\mu} + \tau^*\nabla_\mu\tau, \\ \gamma\tau^* &= \mathcal{L}_\xi\tau^*. \end{aligned}$$

The Lie-prolonged cotangent rows carry the opposite formal-adjoint signs. No coefficient is fitted to a nilpotency defect.

On all twenty-four strict-plus-compensator atoms the independent replay finds $\delta^2 = 0$, $\delta\gamma+\gamma\delta = 0$, and $Q^2 = 0$. It also checks every ghost number, antifield number, parity, and tensor component type.

4.2 Dressed canonical coordinates

The change

$$\widehat{g} = e^{-2\tau} g, qquad\widehat{g}^* = e^{2\tau} g^*, qquad\widehat{\tau}^* = \tau^* + 2g_{\mu\nu} g^{*\mu\nu}$$

obeys the canonical one-form identity

$$g^*\delta g + \tau^*\delta\tau = \widehat{g}^*\delta\widehat{g} + \widehat{\tau}^*\delta\tau.$$

It sends the Weyl Noether row to $\delta\omega^* = \widehat{\tau}^*$ and yields the quartet matrices in Equation 1.

The coordinate change is interpreted in the formal tau-adic local analytic completion. Exact inverse series are checked through order eight, and the all-orders coefficient convolution is

$$\sum_{k=0}^{n} \frac{(-2)^k}{k!} \frac{2^{n-k}}{(n-k)!} = \frac{2^n}{n!}(1-1)^n = \delta_{n0}.$$

5 Extended quotient and restoration

Quartet contraction reduces the extended complex to pure Diff BV cohomology in $\widehat{g}$. The dimension-four ghost-zero basis therefore contains the new invariant $R(\widehat{g})^2$ in addition to $C(\widehat{g})^2$, E_4, and the odd Pontryagin density. $\Box R(\widehat{g})$ remains horizontally exact.

The only dimension-zero natural symmetric tensor that could generate an incoming positive-antifield boundary is $\widehat{g}$ itself. Its image is the trace of the Bach Euler row and vanishes. Normalized coordinate duals for all four surviving ghost-zero classes annihilate both the horizontal and positive-antifield boundary spaces.

At ghost number one the primitives $(B_C, B_E, B_P, B_\Box)$ map to the complete strict four-candidate carrier by the identity matrix in Equation 2. The independently certified pure-Diff anomaly quotient is zero. Hence the full extended dimension-four $H^{1,4}$ vanishes in both parities.

The coefficient-bearing counterterm is

$$S^{(1)}_{\mathrm{ct}} = -\frac{\hbar}{(4\pi)^2}\left(\frac{199}{30} B_C - \frac{87}{20} B_E\right),$$

and exact coordinate replay gives

$$\mathcal{A}^{(1)} + s_{\mathrm{ext}} S^{(1)}_{\mathrm{ct}} = 0 \pmod{\mathrm{d}_h}.$$

This is the one-loop local Euclidean QME restoration theorem in the declared formal tau-adic theory.

6 First Slavnov operator: fixed part and exact ambiguity

The counterterm fixes the local Hamiltonian contribution

$$Q_1^{\mathrm{WZ}} = (C_1^{\mathrm{WZ}}, \cdot).$$

Since C_1^{WZ} has antifield number zero, its field and ghost rows vanish and its cotangent rows are the Euler derivatives with respect to $\widehat{g}$ and τ. This is an actual coefficient-bearing part of Q_1, but not the complete renormalized operator.

The exact non-uniqueness witness uses flat Euclidean momentum $p = (1, 0, 0, 0)$, a TT polarization $h_{11} = 1, h_{22} = -1$, and the conformal polarization $h_{\mu\nu} = \delta_{\mu\nu}$. Direct contraction of the linearized Riemann tensor gives

$$\begin{array}{c|cc} & C^{(1)2} & R^{(1)2} \\ \hline \mathrm{TT} & 1 & 0 \\ \mathrm{conformal} & 0 & 9 \end{array}.$$

Including the topological Euler and Pontryagin columns, the allowed extended $H^{0,4}$ response matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \end{pmatrix}, \qquad \mathrm{rank} = 2.$$

The topological columns have zero compactly supported bulk Euler derivative; their boundary and global content is retained separately. Therefore finite $C(\widehat{g})^2$ and $R(\widehat{g})^2$ counterterms change the bulk Hamiltonian vector field independently. This receipt alone does not supply a nonlocal Γ_1 representative, the renormalized BV Laplacian or time-ordered product, or finite normalization conditions. The follow-up below supplies the anomaly-induced component but not the Weyl-invariant finite remainder. Complete Q_1 remains fail-closed as `NO_CERTIFIED_OPERATOR`.

6.1 Anomaly-induced nonlocal representative

The follow-up receipt solves the exact Weyl-response system rather than declaring the whole finite effective action absent. With

$$\mathcal{E}_4 = E_4 - \frac{2}{3}\Box R, \qquad \Delta_4 = \Box^2 + 2R^{\mu\nu}\nabla_\mu\nabla_\nu - \frac{2}{3}R\Box + \frac{1}{3}(\nabla^\mu R)\nabla_\mu,$$

the carrier basis

$$\left(\langle \mathcal{E}_4, G_4 C^2\rangle, \langle \mathcal{E}_4, G_4 \mathcal{E}_4\rangle, \int \sqrt{g} R^2\right)$$

has Weyl-response matrix diag$(4, 8, -12)$. Exact rational elimination against the modified-basis target $(199/30, -87/20, -29/10)$ gives

$$\left(\frac{199}{120}, -\frac{87}{160}, \frac{29}{120}\right).$$

Re-expansion in $(C^2, E_4, \Box R)$ returns $(199/30, -87/20, 0)$ exactly. Thus

$$\Gamma_{1,\mathrm{anom}} = \frac{1}{(4\pi)^2}\left\{\frac{1}{8}\langle \mathcal{E}_4, G_4(2cC^2 - a\mathcal{E}_4)\rangle + \frac{a}{18}\int \sqrt{g} R^2\right\}$$

is one anomaly-induced Euclidean representative.

The inverse contract is fail-closed. Either the Euclidean boundary problem is invertible, or G_4 is a self-adjoint generalized inverse and every displayed source lies in the complement of $\ker \Delta_4$. A compact Paneitz zero mode is not silently discarded: its global anomaly, homogeneous solutions, and boundary transgressions remain separate data. The Weyl-invariant finite functional, finite C^2/R^2 normalization, and renormalized BV product are still absent, so this result does not promote complete Γ_1 or Q_1.

7 Independent verifiers and mutation tests

The producer and verifier paths are distinct. The verifiers independently:

- validate strict Draft 2020–12 schemas;
- replay exact matrix ranks and contraction identities;
- compare generated certificates with fresh deterministic builds;
- reject deletion of the dressed R^2 class;
- reject authorization of residual transfer;
- reject promotion of the fixed Wess–Zumino contribution to a complete renormalized Q_1;
- replay the diagonal Paneitz/Riegert response solve and reject promotion of a generalized inverse to an unconditional global inverse;
- reject changes to the cotangent signs, gradings, or quartet identity;
- validate every content hash used by the paper claim map.

The generated tables add a publication-layer drift guard. Their verifier loads the eight authoritative table inputs, checks the exact extended boundary matrix and coefficient image, rebuilds the TeX byte-for-byte, and rejects a stale include.

8 Reproduction commands

Run from the repository root:

```
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.wess_zumino_minimal_bv_cotangent_lift --check
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_minimal_bv_cotangent_lift
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.wess_zumino_extended_local_bv_cohomology --check
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_extended_local_bv_cohomology
PYTHONPATH=quantum-weyl python3 -m \
  transfer.one_loop_slavnov_q1_disposition --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_one_loop_slavnov_q1_disposition
python3 paper/generate_12_quantum_anomaly_tables.py --check
python3 paper/verify_12_quantum_anomaly_tables.py
python3 paper/generate_12_pure_weyl_one_loop_bv_anomaly_claim_map.py --check
python3 paper/verify_12_pure_weyl_one_loop_bv_anomaly_claim_map.py
```

The intentionally slow repository-wide suite is not required for this scoped artifact. The relevant exact unit modules, independent verifiers, atlas validator, and two-pass LaTeX build form the declared publication tier.

9 Remaining gate

The next theorem requires actual new physical input:

1. a compensator-inclusive classical contraction $(\iota_{\mathrm{ext}}, \pi_{\mathrm{ext}}, S_{\mathrm{ext}})$;
2. finite normalization conditions, the Weyl-invariant finite nonlocal remainder, global Paneitz Green data, and renormalized BV operator data fixing a complete Q_1 beyond its certified local Wess–Zumino and anomaly-induced contributions;
3. the induced residual operator $q_1 = \pi_{\mathrm{ext}} Q_1 \iota_{\mathrm{ext}}$;
4. nilpotency, adjacent-cohomology, Cartan, mixing, and pairing checks.

Until these land, residual transfer is forbidden. The same-background Hadamard/particle bridge and the classical-obstruction-to-interacting-BRST bridge remain independently fail-closed.